

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input Form was flagged LOWER priority in elicitation, and the benchmark aligns well on this dimension. DrBenchmark is exclusively text-based in standard written French [\[Q31\]](#), implemented via HuggingFace Datasets, Transformers, and PyTorch [\[Q48\]](#), matching the deployment’s text-only clinical NLP pipeline. The Latin alphabet with French diacritics is the natural encoding for both benchmark and deployment, with no script mismatch (QUAERO-D9, MANTRAGSC-D4). Tokenization behavior is investigated through average sub-token counts per word [\[Q75, Q89\]](#), and document splitting at sentence boundaries [\[Q37\]](#) mirrors typical preprocessing. Minor concerns: (1) DEFT-2020 contains a single Japanese-script example (DEFT2020-D13) which is an isolated artifact; (2) some PxCORPUS examples contain ASR artifacts, garbled transcriptions, and embedded system prompts that mix register and form (PXCORPUS-D3, PXCORPUS-D22, PXCORPUS-D27); (3) CLISTER includes degenerate tabular-row inputs (CLISTER-D17) and OCR/PDF extraction artifacts like mid-word line breaks (CLISTER-D18); (4) E3C contains single-punctuation tokenization artifacts in Basque configs (E3C-D15, E3C-D16). None of these compromises the fundamental form alignment for the deployment.

ID	Evidence
Q31	“While the dataset covers 5 languages, only the French portion is retained for the benchmark.” (p.3)
Q37	“Due to the presence of nested entities in annotations, we simplified the evaluation process by retaining only annotations at the higher granularity level from the BigBio (Fries et al., 2022) implementation, following the approach described in Touchent et al. (2023), which translates into an average loss of 6.06% of the annotations on EMEA and 8.90% on MEDLINE. Additionally, considering that some documents from EMEA exceed the maximum input sequence length that most current language models can handle, we decided to split these documents into sentences.” (p.3)
Q48	“We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries.” (p.5)
Q75	“Tokenizers play a crucial role in MLMs by utilizing size-limited vocabularies to split texts into sub-units, aiming to handle out-of-vocabulary (OOV) words.” (p.8)
Q89	“Firstly, we investigated the influence of tokenizers based on the average number of sub-tokens they produce per word.” (p.9)
PXCORPUS-D3	/chet — Single garbled transcription artifact labeled 'none'
MANTRAGSC-D4	Des symptômes potentiellement liés à l' histamine tels que éruption cutanée étendue, gonflement du visage et/ ou des lèvres, démangeaisons, sensation de chaleur ou difficulté à respirer, ont été rapportés. — Well-formed regulatory French with dense symptom NER; confirms formal register alignment
QUAERO-D9	Une cirrhose du foie peut également affecter l'excrétion rénale de la lépirudine. — Standard written French medical prose; confirms script/register alignment
DEFT2020-D13	42 Jeux indémodables (だれでもアソビ大全 , Daredemo Asobi Taizen au Japon...) — Japanese script embedded in French text; video game content; off-domain
E3C-D15	"." tokens=['.'] ner_tags=[0] — Single-punctuation example — tokenization artifact, quality concern
E3C-D16	"." tokens=['.'] ner_tags=[0] — Second single-punctuation example — repeated artifact
CLISTER-D17	Jour 45 - - - - - 25 µg 0,4 mg — Table row, not a clinical sentence; degenerate STS input
CLISTER-D18	Sheldon).\n\nLe patient subissait une chi miothérapie par Méthotrexate - Vinblastine - Endoxan -Cisplatine par voie — Line-break mid-word artifact ("chi miothérapie"); OCR/PDF extraction noise in source material

PXCORPUS-D 22	teralithe 250 milligrammes / le serveur de dialogue met beaucoup de temps à comprendre votre énoncé veuillez reformuler différemment teralithe 250 milligrammes 2 comprimés matin et soir pendant 1 mois — System prompt embedded in prescription utterance; corrupts register
PXCORPUS-D 27	tropatopine chlorhydrate 10 milligrammes 1 comprimé le midi — "1o" instead of "10" — ASR/OCR artifact in dose quantity

Strengths

- Exclusively text-only standard French aligned to deployment's text-only pipeline [Q31, Q48]
- Latin alphabet with French diacritics natively supported throughout (QUAERO-D9, MANTRAGSC-D4, DEFT2021-D1)
- HuggingFace/PyTorch implementation provides reproducible toolkit matching modern French biomedical NLP practice [Q48]
- Tokenization investigated empirically across 8 MLMs [Q75, Q78, Q89]

ID	Evidence
Q31	"While the dataset covers 5 languages, only the French portion is retained for the benchmark." (p.3)
Q48	"We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries." (p.5)
Q75	"Tokenizers play a crucial role in MLMs by utilizing size-limited vocabularies to split texts into sub-units, aiming to handle out-of-vocabulary (OOV) words." (p.8)
Q78	"However, our experiments, as shown in Table 7, reveal that FlauBERT is the model with the least word segmentation (1.43 in average), while DrBERT-CP tends to have the highest average segmentation (1.90 in average)." (p.8)
Q89	"Firstly, we investigated the influence of tokenizers based on the average number of sub-tokens they produce per word." (p.9)
DEFT2021-D1	Un homme de 42 ans était hospitalisé en urgence le 10 décembre 2003 dans le service de gastro-entérologie pour un tableau d'hépatite aiguë — Multi-morbid hepatology/toxicology/psychiatry case with 6 concurrent MeSH labels
MANTRAGSC-D4	Des symptômes potentiellement liés à l'histamine tels que éruption cutanée étendue, gonflement du visage et/ ou des lèvres, démangeaisons, sensation de chaleur ou difficulté à respirer, ont été rapportés. — Well-formed regulatory French with dense symptom NER; confirms formal register alignment
QUAERO-D9	Une cirrhose du foie peut également affecter l'excrétion rénale de la lépirudine. — Standard written French medical prose; confirms script/register alignment

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both benchmark and deployment are text-only, standard written French with French diacritics. No mismatch in signal encoding.

IF-2: Regional infrastructure (French hospital IT) supports the same UTF-8 / Latin-diacritical text encoding. HDS-certified hosting requirements [WEB-4] govern data hosting but do not affect signal form.

IF-3: Domain-specific form: tokenization of medical compounds and abbreviations. DrBenchmark studies sub-token counts across 8 models [Q75, Q78] and finds tokenization plays a minor role in performance [Q80].

IF-4: Minor form mismatches: DEFT-2020 contains incidental Japanese script (DEFT2020-D13); PxCorpus contains ASR artifacts and embedded system prompts (PXCORPUS-D3, PXCORPUS-D22); CLISTER contains OCR/PDF extraction noise (CLISTER-D18). These are isolated and do not threaten external validity for the deployment.

ID	Evidence
Q31	"While the dataset covers 5 languages, only the French portion is retained for the benchmark." (p.3)
Q48	"We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries." (p.5)
Q75	"Tokenizers play a crucial role in MLMs by utilizing size-limited vocabularies to split texts into sub-units, aiming to handle out-of-vocabulary (OOV) words." (p.8)
Q78	"However, our experiments, as shown in Table 7, reveal that FlauBERT is the model with the least word segmentation (1.43 in average), while DrBERT-CP tends to have the highest average segmentation (1.90 in average)." (p.8)
Q80	"Yet, tokenization, as it is currently done in MLMs, seems to play a minor role in the performance of systems." (p.8)
PXCORPUS-D3	/chet — Single garbled transcription artifact labeled 'none'
QUAERO-D9	Une cirrhose du foie peut également affecter l'excrétion rénale de la lépirudine. — Standard written French medical prose; confirms script/register alignment
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PXCORPUS-D22	teralithe 250 milligrammes / le serveur de dialogue met beaucoup de temps à comprendre votre énoncé veuillez reformuler différemment teralithe 250 milligrammes 2 comprimés matin et soir pendant 1 mois — System prompt embedded in prescription utterance; corrupts register
WEB-4	https://engage.hoganlovells.com/knowledgeservices/news/health-data-hosting-the-new-french-hds-certification-has-been-released